Learning mixture thermodynamics with efficient screening adaptation and thermodynamically guided replay

Daniel Morales Salazar

Independent Researcher, Sweden

Abstract

Screening multicomponent liquids requires economical thermodynamic prediction, reliable instability detection, and adaptation to new composition regimes without losing source-domain performance. We study water, methanol, acetone, acetonitrile, and benzene at 298.15 K through a connected sequence of regression, classification, active learning, transfer adaptation, and replay experiments. Molecular structures were optimized with ORCA 6.1.1 at BP86/def2-TZVP in a conductor environment, followed by BP86/def2-TZVPD surface calculations for openCOSMO-RS 24a. Equimolar calculations illustrated how favorable ideal mixing can be offset by nonideal interactions. A composition-only fourth-degree polynomial Ridge model predicted excess Gibbs energy, with the analytical ideal contribution restored to obtain total mixing energy. Held-out mean absolute error was 0.00990 J/mol, compared with 13.397 J/mol for histogram gradient boosting; Ridge remained accurate under modest extrapolation (0.195 J/mol), but its error reached 115.357 J/mol on a water–benzene binary challenge. These errors measure agreement with the computational reference. For instability classification, 2,000 recipes with all five solvents present at 15–40 mol% each were labeled by a 40-start tangent-plane search with verification of energy-lowering splits. Composition-only and molecular-descriptor MLP committees reached average precision (AP) of 0.9898 and 0.9974; activity-based and full-feature committees reached 1.0000. Feed activities require one-temperature evaluation and avoid the multistart search. Across five seeds, source active learning reached mean AP 0.9930 with 200 labels, versus 0.9250 for random selection and 0.9928 for a matched 1,191-label reference. Frozen models generalized usefully to broader five-component compositions outside the original 15–40 mol% bounds but made confident binary-limit errors. Transfer-active adaptation of P3 active-200 reached target AP 0.9534 with 150 target labels while source AP fell to 0.6698, motivating 20-example replay memories guided by historical search duration or predicted mixing energy. In the larger P2 molecular setting, the strict energy bridge spanning the first label transition in the energy-ranked source pool achieved source AP 0.9518 and 0.9452 at target budgets 150 and 300, exceeding balanced random controls by 0.1612 and 0.1415; four of five adaptation seeds favored the bridge at each budget. This retention came with lower target AP than the balanced control. Matched scratch-initialization controls further supported beneficial transfer from source pretraining, with five-seed mean target AP of 0.9312 versus 0.8867 for the P2 strict bridge at

300 target labels, using identical memories, target examples, and source-fitted preprocessing. Duration-guided replay and its positive-enriched class-matched control also supported source retention: their exploratory equal-weight mean source AP exceeded uniform random replay in all six source-family/budget combinations. Selecting the longest-running examples added uneven source-AP gains beyond class matching, while improving P3 target F1 over the class-matched control in all five runs at 150 labels. The strict bridge required ten positive examples immediately above the first negative in descending predicted-energy order and ten negatives from that transition downward; it was feasible for only one active-200 seed. A downstream bridge extension preserved source ranking across five seeds and achieved higher mean source F1 than its balanced control at all three budgets, although its source AP advantages were mixed. The results support economical screening and useful, bounded extrapolation, while identifying thermodynamically guided replay, historical search duration, and the class allocation induced by that duration ranking as useful ways to construct retention memories, with benefits dependent on the source pool and target tradeoff. Within replay-eligible source pools, both long-search and high-energy selection enriched detected instabilities. This links recorded computational effort and thermodynamic ordering to the rare positive class and motivates reusing them as informative signals for replay selection.

Introduction

The composition of a liquid mixture influences solvation, chemical activities, and the tendency to remain homogeneous or separate into phases. Understanding these changes can help identify operating regions for reactions, dissolution, extraction, and purification. A homogeneous liquid can maintain contact between dissolved reactants, whereas phase separation can be useful when products or impurities preferentially partition between phases. An instability map is therefore valuable both for locating regimes to avoid and for identifying regimes that may support a separation. Further information about solutes, equilibrium phase compositions, and partitioning is needed to establish process performance; locating an instability is one component of that broader design problem. COSMO-RS provides an established route from quantum-chemical molecular information to solvent screening and liquid-phase thermodynamics [1].

openCOSMO-RS makes this approach available through an open implementation, including the 24a parameterization used here [2]. Once molecular surface information has been calculated, it can be reused across many compositions of the same components. We selected water, methanol, acetone, acetonitrile, and benzene to combine hydrogen-bonding, polar aprotic, and nonpolar aromatic solvents in one manageable system. Five mole fractions define a four-dimensional composition space because they sum to one. This setting allows us to investigate increasing compositional complexity without simultaneously changing the molecular identities at every machine-learning query. The present study concerns neutral solvent mixtures.

Our starting question was deliberately simple: how do the calculated mixing contributions change when we begin with equal molar amounts and successively add components? Equimolar mixtures provide a transparent reference because the ideal mixing entropy is $R \ln n$ for n components. Their nonideal contributions then show why greater component count alone does not determine the total mixing free energy. The next question was whether a trained model could cheaply reproduce these composition-dependent quantities across a restricted interior region. Polynomial descriptions of mixture thermodynamics have a long history [3]. We used polynomial Ridge regression as a numerical surrogate for the calculated excess Gibbs energy.

A homogeneous liquid has a uniform composition, but its Gibbs energy can still be lowered by separating into liquids of different compositions. Negative mixing Gibbs energy relative to the pure components does not by itself establish stability against such separation. At fixed temperature and pressure, the tangent plane to the molar Gibbs-energy surface at the feed composition provides the relevant comparison: a trial composition below this plane identifies a direction in which forming a new liquid phase can lower the total Gibbs energy [4].

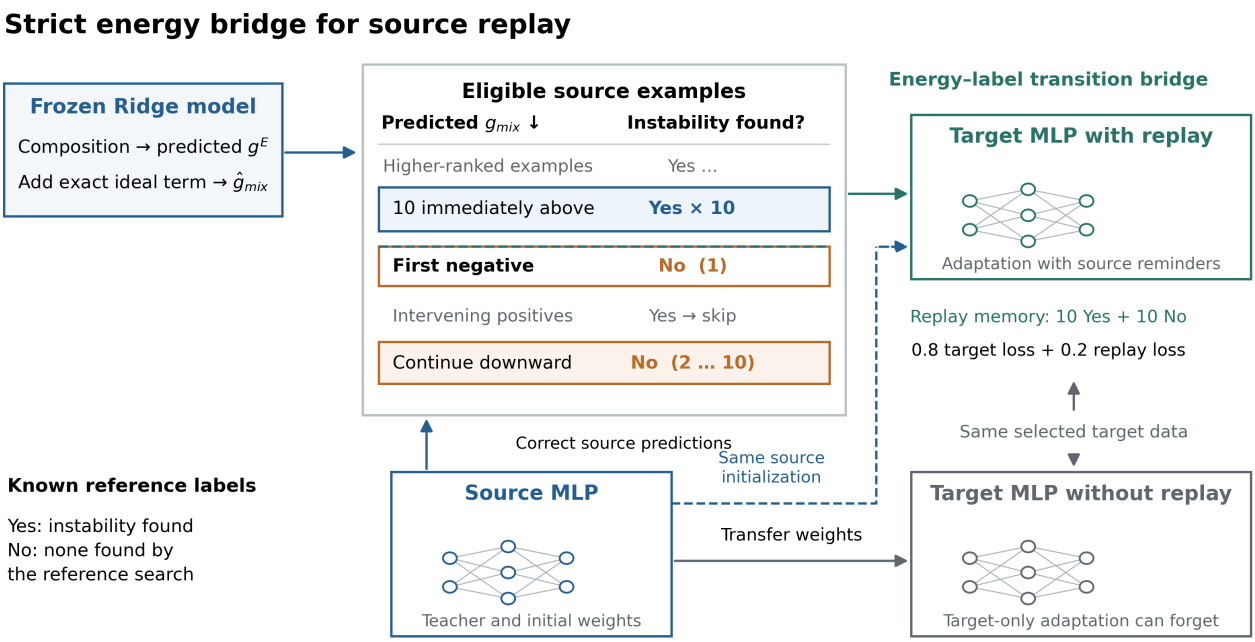
We therefore screened each feed with a multistart tangent-plane search. Sequential least squares programming (SLSQP), a constrained local optimization method, explored trial-phase compositions from 40 starting points, including the feed, component-rich, pair-rich, and random compositions [12]. A negative tangent-plane result was followed by verification of an energy-lowering, mass-balanced two-liquid split. Forty refers to optimization starting points; each local search can require many iterations and thermodynamic evaluations. Methods M3 gives the objective, numerical settings, and split-verification equations.

This difference in cost motivates a hierarchy of predictors. Composition-only models need no new thermodynamic calculation at inference. Simple molecular descriptors and a volume-additive density estimate can also be constructed from known composition and component constants. Activity coefficients at the feed require one temperature calculation, while finite-difference entropy and enthalpy require neighboring temperatures as well. Even the richer feed features can remain attractive relative to the full stability search. Learning activity coefficients and multicomponent interactions has also been investigated using molecular graph models [5]; our fixed-solvent feature comparisons address a narrower question about economical prediction within a known chemical system.

Training-data acquisition introduces another opportunity to save resources. Active learning selects which compositions should receive expensive reference labels, while transfer learning reuses information learned in one domain when fitting another [6,7]. We first learned within a restricted five-component interior region, then examined progressively broader compositions and lower-dimensional faces of the simplex. This is an extrapolation problem in composition space even though the solvent identities remain unchanged. We expected that nearby departures might remain tractable, but treated behavior near dilute limits, missing components, and pure endpoints as an empirical question. Subsequent target adaptation was

evaluated against random acquisition, scratch training, and scratch training on the exact same selected target examples.

Finally, adaptation raised a retention problem: improving predictions in new regimes could damage predictions in the original region. This deterioration of previously learned performance during subsequent training is widely known as catastrophic forgetting [10]. Small replay memories offer a practical way to preserve previous knowledge [8]. We asked whether the thermodynamic workflow itself could help choose useful reminders. Long historical stability-search durations might identify examples worth retaining, while an energy-ranked memory could preserve contrasting labels near a transition in the source data. The energy bridge was developed to retain examples from both sides of the first transition encountered when moving from high to low predicted mixing free energy. Both strategies test whether information already produced by the thermodynamic workflow improves retention beyond replay size and class balance. Duration-guided selection additionally tests whether examples associated with expensive searches provide useful reminders; an explicit relationship between search duration and intrinsic ‘thermodynamic difficulty’ and computational complexity remains a hypothesis. Scheme 1 summarizes the strict bridge and its role during adaptation. The following account links these questions, the controls used to answer them, and the resource costs incurred at each stage. The Methods section specifies the calculation and learning procedures; Appendix A collects detailed settings, and Appendix B states code and data availability.



Scheme 1. Strict energy-bridge selection and replay. The frozen Ridge predictor supplies excess Gibbs energy, to which the exact ideal contribution is added. Source-training examples correctly classified by the frozen source MLP are sorted by predicted total mixing Gibbs energy. The memory contains the ten positives immediately preceding the first negative and the first ten negatives encountered from that anchor downward, skipping any intervening positives. Yes and No are previously obtained reference labels, not labels assigned by Ridge. Replay and no-replay fits start from the same source model and use the same selected target data; replay contributes 20% of the domain-weighted loss. The schematic illustrates the strict rule used in P2; the relaxed P3 extension is described later in Section 11. Retention benefits and target tradeoffs are evaluated in Section 11.

Results and Discussion

1. Establishing the molecular and thermodynamic calculations

We first established the thermodynamic workflow with available reference molecular surfaces, then generated a consistent set for our five molecules using ORCA 6.1.1 and the quantum-chemical levels used for openCOSMO-RS 24a [2,9]. The resulting surfaces supported calculation of activity coefficients, excess and total mixing Gibbs energies, and finite-difference mixing entropies and enthalpies at 298.15 K. Molecular preparation, standard states, and thermodynamic equations are specified in Methods M1.

For the equimolar cases, changing the temperature-difference step from 1 to 2 K changed the estimated enthalpy by approximately 0.000017–0.000062 kJ/mol. This checks numerical differentiation within the chosen model. We also constructed volume-additive molar-volume and density descriptors for the later learning experiments (Methods M2).

2. Beginning with equimolar mixtures

We started with equimolar fractions: water–methanol at 0.5 each, followed by three-, four-, and five-component mixtures formed by adding acetone, acetonitrile, and benzene in that order. At each stage, all components present had equal mole fractions. We chose this sequence as a simple reference for comparing ideal and excess contributions. For a fixed number of components, equimolar composition maximizes the ideal mixing entropy, but it need not minimize the total Gibbs energy or yield a stable homogeneous liquid. Table 1 reports the calculations using our independently generated molecular surfaces.

Table 1. Equimolar thermodynamics from the calculated molecular surfaces at 298.15 K. Gibbs energies and enthalpy are in kJ/mol; mixing entropy is in J/(mol K).

Components added cumulatively	Ideal Gibbs energy	Excess Gibbs energy	Total mixing Gibbs energy	Mixing enthalpy	Mixing entropy
Water and methanol	−1.7183	0.4465	−1.2718	−0.3972	2.9334
Plus acetone	−2.7234	0.2831	−2.4403	−1.1673	4.2698
Plus acetonitrile	−3.4366	0.6100	−2.8265	−0.1745	8.8951
Plus benzene	−3.9897	1.1464	−2.8433	0.2395	10.3397

The ideal contribution became more favorable with each added component, but the excess contribution did not follow a monotonic favorable trend. The clearest example was the change from four to five components: an additional ideal contribution of −0.5532 kJ/mol was almost canceled by a +0.5364 kJ/mol increase in excess Gibbs energy. The total changed by only −0.0168 kJ/mol. The five-component mixture had positive mixing enthalpy, while its favorable entropic contribution kept total mixing Gibbs energy negative. The preliminary reference-surface calculations gave the same qualitative picture, including the near cancellation on adding benzene.

Motivated by our interest in high-entropy electrolytes, we used these preliminary neutral-mixture calculations to build an understanding of how ideal entropy and nonideal interactions compete as the solvent composition changes. This provides a starting point for later electrolyte studies that explicitly include salts and their interactions.

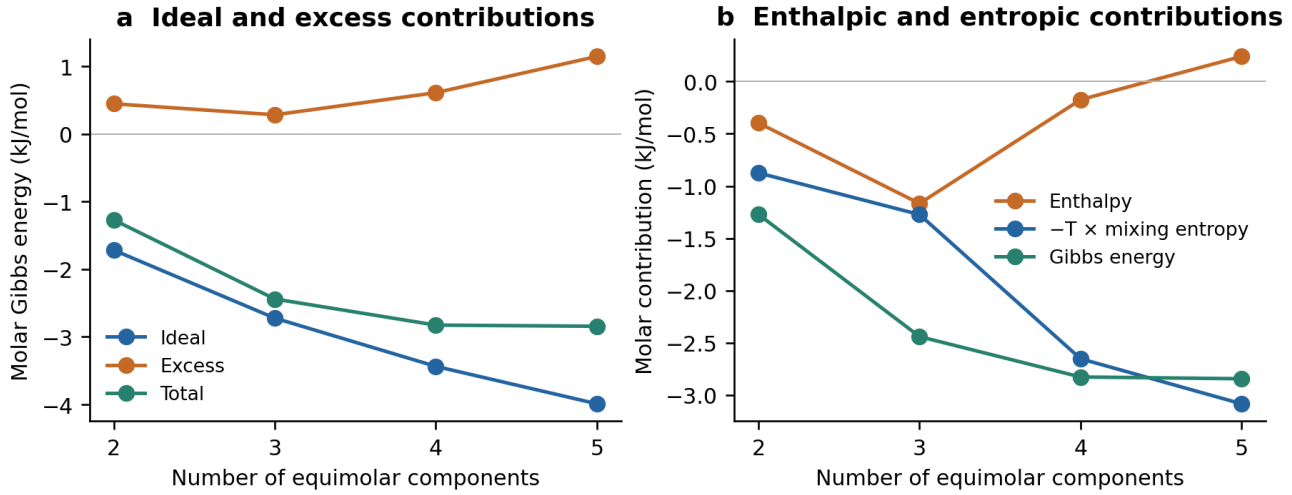


Figure 1. Thermodynamic contributions along the equimolar sequence. Panel a separates ideal and excess Gibbs energy; panel b separates enthalpic and entropic contributions. Both solvent identity and composition change along this sequence, so it does not isolate an effect of component count alone.

These observations led to a practical question: would the five solvents remain mutually miscible throughout an interior composition region in which each component contributes 15–40 mol%, or could the reference model predict liquid–liquid separation even there? Neither the component count nor a negative mixing Gibbs energy answers that question. We therefore sought both a broader energy map and a direct instability screen. Repeated calculations over this four-dimensional composition space also motivated an inexpensive surrogate.

3. Project 1 (P1), learning excess and total Gibbs mixing energies

We constructed 23,751 compositions in the bounded source region, with each solvent at 15–40 mol%, and reserved disjoint composition blocks for training, validation, and testing. Four independent mole fractions defined each input; the benzene fraction followed from their sum (Methods M2).

The learned quantity, excess Gibbs energy, is the difference between the molar Gibbs energy of the actual solution and that of an ideal solution at the same temperature, pressure, and composition. Total mixing Gibbs energy instead describes the change relative to the separated pure components and equals the ideal mixing contribution plus the excess contribution. Excess Gibbs energy therefore measures nonideality [11, Chapter 6].

We trained the surrogate on excess Gibbs energy and reconstructed total mixing Gibbs energy by adding the analytical ideal contribution.

We compared a training-mean baseline, polynomial Ridge regression, and histogram gradient boosting across training sizes from 100 to 17,121 recipes. Validation selected degree-4 Ridge with regularization 0.0001 and boosting with 31 maximum leaf nodes and learning rate 0.1. The complete selection grid and frozen-test procedure are given in Methods M4 and Appendix A1.

The composition-only Ridge model reproduced the computational energy surface with very small errors. Its test mean absolute error was 0.000009903 kJ/mol, or 0.009903 J/mol, compared with 0.013397 kJ/mol for boosting and 0.116726 kJ/mol for the mean-excess baseline. Total-energy R^2 was approximately 0.999999989 for Ridge. Because the ideal contribution was restored exactly, the excess- and total-energy absolute errors were identical. The result shows that a smooth, restricted composition-dependent energy surface can be represented extremely well with a modest polynomial model. These errors measure agreement with the openCOSMO-RS reference [1,2].

We then froze the regressors and evaluated fresh off-grid source-region recipes, modest and broader departures from the original composition bounds, and a 111-point water–benzene challenge including dilute and pure limits. Each five-component group contained 200 recipes (Methods M4). Table 2 shows how predictive usefulness changed with the extent and type of extrapolation.

Table 2. Frozen Gibbs-energy regression across domains. Errors are in J/mol.

Evaluation group	Recipes	Ridge MAE	Boosting MAE	Ridge maximum absolute error
Blocked in-domain test	3,758	0.00990	13.397	0.056
Fresh off-grid in domain	200	0.01132	13.936	0.073
Mild OOD	200	0.19537	65.109	3.402
Broad OOD	200	3.76682	243.502	94.728
Water–benzene binary challenge	111	115.357	1,073.057	611.837

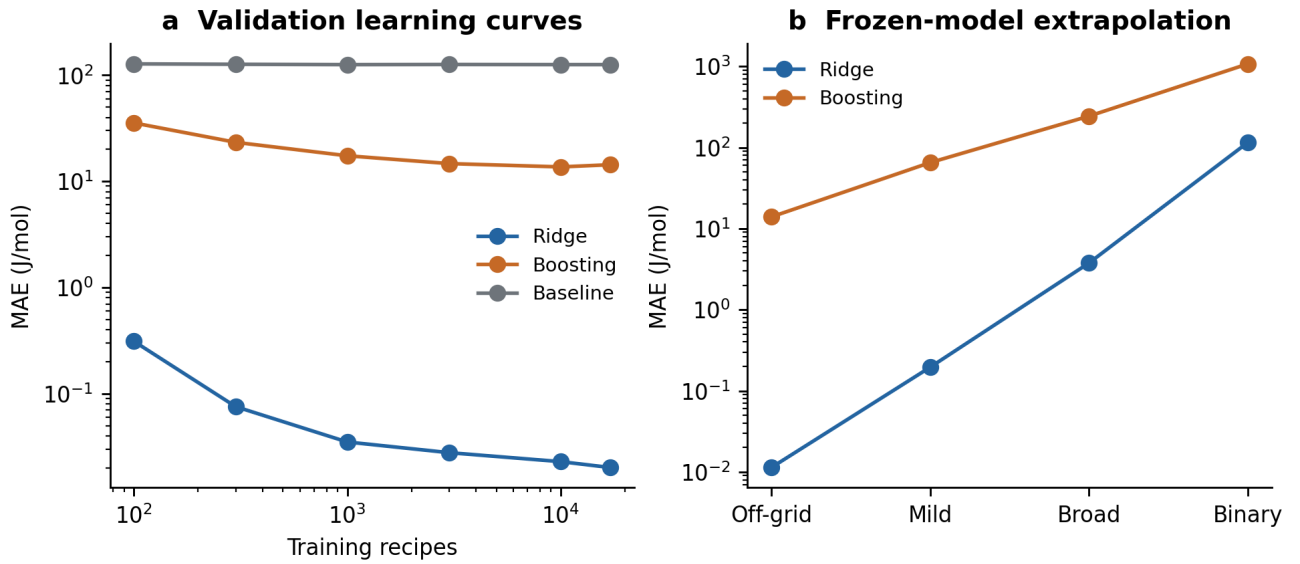


Figure 2. Polynomial Ridge and boosting performance. Panel a shows validation MAE at each training size, using the validation-selected setting within each family. Panel b shows frozen-model MAE on the extrapolation groups. Logarithmic error axes reveal the large differences in absolute scale; these groups also differ in composition coverage and size.

The useful finding was a progression in reliability. Fresh off-grid error remained close to the original test error, and mild extrapolation remained highly accurate against openCOSMO-RS (Table 2). Even broad OOD had a small mean error relative to the scale of the total energy, although its worst case was materially larger (94.728 J/mol maximum absolute error versus 3.767 J/mol MAE). The binary challenge exposed the anticipated limitation of extrapolating far from the training region: Ridge MAE rose to 0.1154 kJ/mol, with a maximum error of 0.6118 kJ/mol. At the exact pure-water and pure-benzene endpoints, the unconstrained Ridge polynomial returned mixing free energies of +0.1810 and +0.6118 kJ/mol, respectively, although the reference value is zero. These endpoints lay outside training, and the condition $g_{mix}=0$ at pure-component limits had not been imposed on the fit. Boosting was less accurate in every reported group and substantially worse at the binary limit. The result supports a useful local surrogate with some extrapolation range, while identifying the boundary behavior that an unconstrained polynomial does not reliably recover.

This distinction became important later. We reused the frozen Ridge model to rank source-region replay candidates, where its approximation was well supported.

4. Project 2 (P2), learning the outcome of a stability search

The next question was whether a classifier could identify recipes for which the computational search detects instability. The search asks whether a homogeneous mixture can lower its Gibbs energy by separating into liquids with different component ratios. We label a recipe as a detected instability when a negative tangent-plane candidate yields a mass-balanced split with lower Gibbs energy than the original mixture.

We began with 600 source-region recipes: all 126 points of a coarser 5 mol% grid plus 474 sampled recipes from the finer grid. The initial screen found 47 positives. We expanded the dataset by 1,400 additional sampled recipes to obtain 2,000 labeled compositions, with 154 detected instabilities. The final split contained 1,191 training recipes with 91 positives, 314 validation recipes with 30 positives, and 495 test recipes with 33 positives.

Reference labels came from a multistart tangent-plane search with openCOSMO-RS activities [4]. A positive required an explicitly verified energy-lowering, mass-balanced split. A negative required no instability detected within the finite 40-start search and convergence of all its local searches; unresolved cases were excluded. Our obtained outputs are computational search outcomes, not proofs of global stability. Methods M3 gives the objective, tolerances, and verification procedure.

We compared a majority baseline, logistic regression, an RBF support-vector classifier, and three-member MLP committees with composition, molecular, thermodynamic, activity, and full feature groups. Composition-only and molecular-descriptor committees are termed P2 composition and P2 molecular, respectively. Grouped training-only model selection and validation-selected cutoffs were shared across this comparison (Methods M5).

Table 3. Test performance of the instability classifiers. Each row uses the same 495 recipes and its validation-selected cutoff. MLP denotes a three-member committee. FP and FN are false-positive and false-negative counts.

Inputs and learner	AP	Recall	F1	FP	FN
Composition logistic	0.9148	0.9697	0.7711	18	1
Composition RBF SVM	0.9729	0.8485	0.9180	0	5
Composition MLP	0.9898	0.9697	0.9275	4	1
Molecular MLP	0.9974	0.9394	0.9688	0	2
Thermodynamic MLP	0.9960	0.9697	0.9143	5	1
Activity logistic	0.9951	0.9697	0.9275	4	1
Activity MLP	1.0000	0.9394	0.9688	0	2
Full-feature MLP	1.0000	0.9394	0.9688	0	2

On this test set, P2 molecular slightly exceeded P2 composition in average precision (AP) and F1: AP increased from 0.9898 to 0.9974, and F1 from 0.9275 to 0.9688. This improvement came with one additional missed positive but four fewer false alarms. Consequently, “better” depends partly on the intended screening decision. For this fixed solvent set, the molecular descriptors are deterministic functions of composition and known constants; their benefit concerns representation and fitting rather than independent chemistry.

Activity-based and full-feature MLPs gave the best ranking, with AP 1.0000. Both nevertheless missed two positives at their validation-selected cutoffs. Perfect ordering means that a threshold separating the two classes exists for these test scores, but a cutoff selected on a different set need not fall in that separating interval. The reported classifications retain the validation-selected cutoffs. Activity features also helped a simpler learner: logistic AP rose from 0.9148 with composition to 0.9951 with activities, while false positives fell from 18 to four at the same 32/33 recall. Thermodynamic features also improved ranking, although not every learner–feature combination improved F1. The full model matrix is retained in the accompanying data.

The strong activity-based logistic result suggests a useful role for thermodynamically informed features. Activities satisfy $a_i = x_i \gamma_i$, and the corresponding chemical potentials contain $RT \ln a_i$. The supplied feed features, $\ln \gamma_i$, therefore encode nonlinear interaction effects that a composition-only model must learn from the labeled data. Their connection to the tangent-plane criterion helps explain why a linear logistic decision function could rank instability so well in this feature space. The observed improvement is consistent with a boundary that is easier to approximate linearly after this transformation.

Activity-based prediction can also be economical relative to direct instability searching. It requires a feed calculation at one temperature, but avoids temperature differentiation and the multistart phase search. For the 495 source-test recipes, the recorded median all-feature generation time was 0.9105 s per recipe and the median stability-search time was 91.4365 s. The latter was calculated by taking the median of the saved per-recipe stability_seconds

values, each measured around the complete 40-start search, split verification, and associated result recording. It is neither the time per optimization start nor the interval between completed jobs in a parallel run. Both medians are recorded in the benchmark_v1 report's prediction_costs.csv, in the campaign_all_features_median_seconds and campaign_stability_median_seconds columns.

These timings describe the recorded campaign workload. Individual searches and completion intervals can differ substantially, and parallel resource contention can affect per-job elapsed time. The all-feature timing also includes more work than activity-only inference, so it is not an isolated benchmark for that feature group. Nevertheless, the comparison supports spending some computation on feed features when they improve predictions enough to avoid a full search. Batch MLP scoring itself took approximately 2.6–5.2 microseconds per recipe, excluding feature generation and other execution overhead.

The majority classifier illustrates why accuracy alone is insufficient: predicting every recipe as negative gives 93.33% accuracy on this test set while detecting none of the 33 positives. Its constant scores give AP 0.0667, equal to the positive prevalence, and instability recall and F1 are zero. We therefore use AP to assess ranking and interpret it alongside recall, F1, and false-positive counts at the validation-selected cutoff. High AP with poor thresholded performance calls for examining the operating point before judging the model's practical usefulness.

5. Project 3, acquiring informative source labels

Project 3 (P3) asked whether informative acquisition could reproduce source instability labels with less reference work. It used the existing 1,191-recipe source-training pool retrospectively, with validation and test sets fixed. Five seeds compared acquisition policies using three-member composition-only MLP committees (Methods M6).

Random sampling, space filling, and uncertainty plus diversity began with the same 100 labeled recipes within each seed and progressed to common reporting budgets. The active rule combined committee disagreement, prediction ambiguity, composition-space diversity, and random exploration; predictions selected reference queries. We use P3 active-200 and P3 active-800 for the committees trained under this policy at those label budgets. A matched full-pool committee supplied the reference.

Table 4. Source performance at 200 acquired labels and for the full-pool reference. Values are means across five seeds on one shared 495-recipe test set.

Strategy	Training labels	AP	Recall	F1	Recorded reference-duration sum
Random	200	0.9250	0.7212	0.7663	4.694 h
Space filling	200	0.9535	0.6545	0.7304	4.330 h
Uncertainty plus diversity	200	0.9930	0.9394	0.9598	5.130 h
Matched full-pool reference	1,191	0.9928	0.9394	0.9568	27.775 h

Active acquisition reached strong performance at 200 labels: mean AP 0.9930, recall 0.9394, and F1 0.9598. Every active-200 seed exceeded its paired random and space-filling run in AP and F1. All five active trajectories first met the declared AP ≥ 0.98 and recall ≥ 0.90 criterion at the 200-label checkpoint. Random selection first met it at 800 labels for two of five seeds. Space filling first met it at 400 labels for two seeds and at 800 for one; the remaining runs did not meet both criteria at the evaluated checkpoints. The active rule had acquired a mean of 39 positives at 200 labels, compared with 14.8 for random and 16.6 for space filling. This enrichment helps explain its efficiency in a dataset where positives were uncommon.

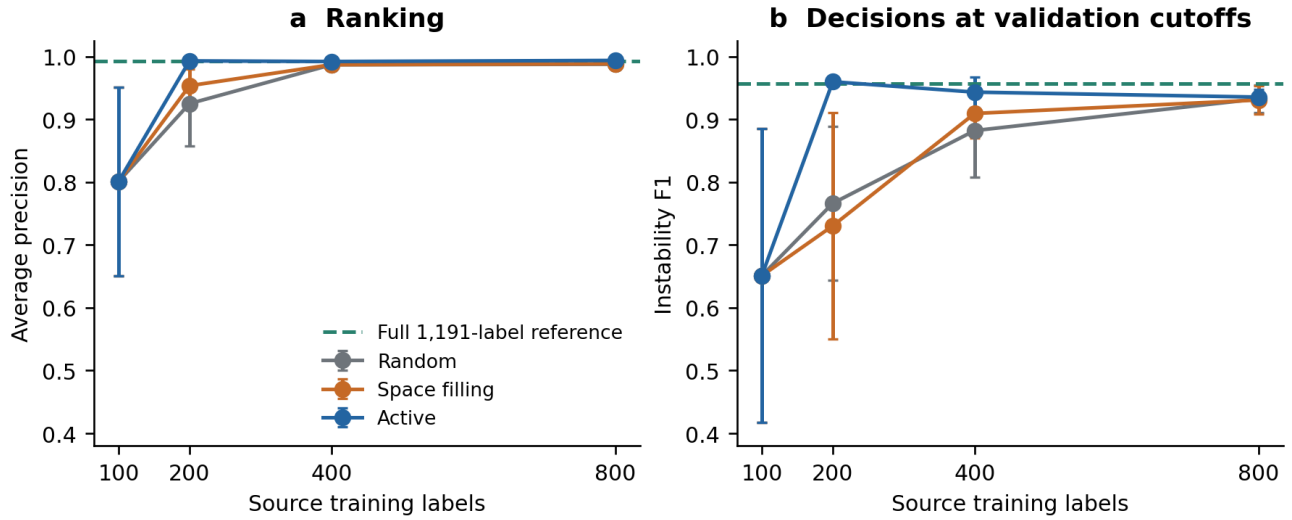


Figure 3. Source active-learning curves. Points are five-seed means and error bars are sample standard deviations across seeds. The dashed line shows the matched full-pool reference. All strategies use the same validation and test sets.

Active-200 reduced the number of required labels while selecting a costlier subset at the same label budget. Its mean summed historical stability-search duration was 5.130 h, compared with 4.694 h for random-200 and 4.330 h for space-filling-200 (Table 4). This coincided with its greater positive-label yield: 39 positives on average, versus 14.8 and 16.6. Together, these observations led us to the hypothesis that recipes requiring longer searches might carry useful information about difficult or unstable regions and could be valuable reminders during later adaptation. This motivated us to run the empirical slowest-20 replay strategy in Section 10. The increase in both positive-label yield and recorded duration provided an empirical motivation for this test.

Active-200's summed recorded reference duration remained 81.5% below that of the full training pool. Its cumulative model-fitting duration was approximately 53.9 s and acquisition time 0.122 s, additional to the reference-duration proxy. Because these labels already existed, the retrospective experiment estimated what a similarly selected future campaign might require.

More labels did not monotonically improve the thresholded scores. Active-800 retained excellent AP, 0.9937, but mean recall fell to 0.8788 and F1 to 0.9354. Its mean false positives

fell from 0.6 at 200 labels to zero, while false negatives increased from two to four. The ranking remained strong, but the validation-selected decisions were more conservative. Recall and F1 depend on the decision cutoff, whereas AP evaluates the ordering of the scores across cutoffs. A cutoff can be retuned after training on independent validation or calibration data to reflect the desired precision–recall tradeoff, without changing the learned weights. Thus, active-800's lower F1 need not imply an irreversible loss of predictive information.

6. Project 4, testing frozen models beyond the source region

Project 4 (P4) investigated generalization beyond the original composition region (this section), adaptation to new composition regimes (Section 7), and preservation of source-domain performance during adaptation (Sections 9–11). We first evaluated the frozen source models on a composition stress test, retaining their saved preprocessing and decision cutoffs without further training. The challenge comprised 335 recipes: 40 fresh off-grid compositions within the original 15–40 mol% per-component bounds, 40 mild-OOD compositions within 10–45 mol % but outside those source bounds, 40 broad-OOD compositions within 1–90 mol% but outside the 10–45 mol% region, 210 binary recipes covering all ten solvent pairs, and five pure endpoints. The bounds apply to each of the five solvents; all fractions sum to one, and leaving a region requires at least one component to violate its bounds. Of the 330 searched recipes, 36 had detected instabilities; the five pure endpoints were analytic negative reference checks. Our challenge tested both departures from the interior bounds and movement onto lower-dimensional faces of the composition simplex (Methods M7).

Table 5. Frozen five-component classification at the saved source cutoffs. Each group contains 40 recipes, with 3, 8, and 10 positive references in the off-grid, mild OOD, and broad OOD groups, respectively. P3 active-200 entries are means across five source seeds; P2 entries describe the saved source committees. FP and FN denote false-positive and false-negative counts.

Group	Model	AP	Recall	F1	FP	FN
Off-grid in domain	P2 composition	1.0000	1.0000	1.0000	0	0
Off-grid in domain	P2 molecular	1.0000	1.0000	1.0000	0	0
Off-grid in domain	P3 active-200	1.0000	1.0000	1.0000	0	0
Mild OOD	P2 composition	1.0000	1.0000	0.9412	1	0
Mild OOD	P2 molecular	1.0000	0.8750	0.9333	0	1
Mild OOD	P3 active-200	0.9941	0.9250	0.9483	0.2	0.6
Broad OOD	P2 composition	0.9833	1.0000	0.9091	2	0
Broad OOD	P2 molecular	0.9833	0.9000	0.8571	2	1
Broad OOD	P3 active-200	0.9792	0.9000	0.8571	2	1

The classifiers showed useful extrapolation on the five-component groups (Table 5). P2 composition missed no positives and produced one false positive in mild OOD and two in broad OOD. P2 molecular missed one positive in each OOD group, with zero and two false

positives, respectively. Leaving the original bounds therefore did not immediately remove predictive usefulness, although these 40-recipe groups were small. Binary behavior was less reliable: water–benzene was the only pair with detected instabilities, yet P2 composition, P2 molecular, and every P3 active-200 committee confidently flagged all 21 water–benzene recipes, including the six negative references nearest the pure limits. These models, trained only on the five-component interior, failed to recover the return to negative reference labels near either end of this binary composition range. One possible explanation is that a learned association between water–benzene-rich compositions and instability extended beyond its valid range. Across all ten binary pairs, the models incorrectly flagged roughly one-third of the 195 negative references (Figure 4b), although methanol–acetone, methanol–acetonitrile, and acetone–acetonitrile were classified correctly throughout. All three model families also flagged two of the five pure endpoints. Increasing P3 source training from 200 to 800 labels did not materially repair these errors: additional examples from the same interior region did not supply the missing boundary behavior. Activity-based and full-feature classifiers were not evaluated in this frozen challenge and ternary and quaternary mixtures entered the subsequent target-adaptation dataset.

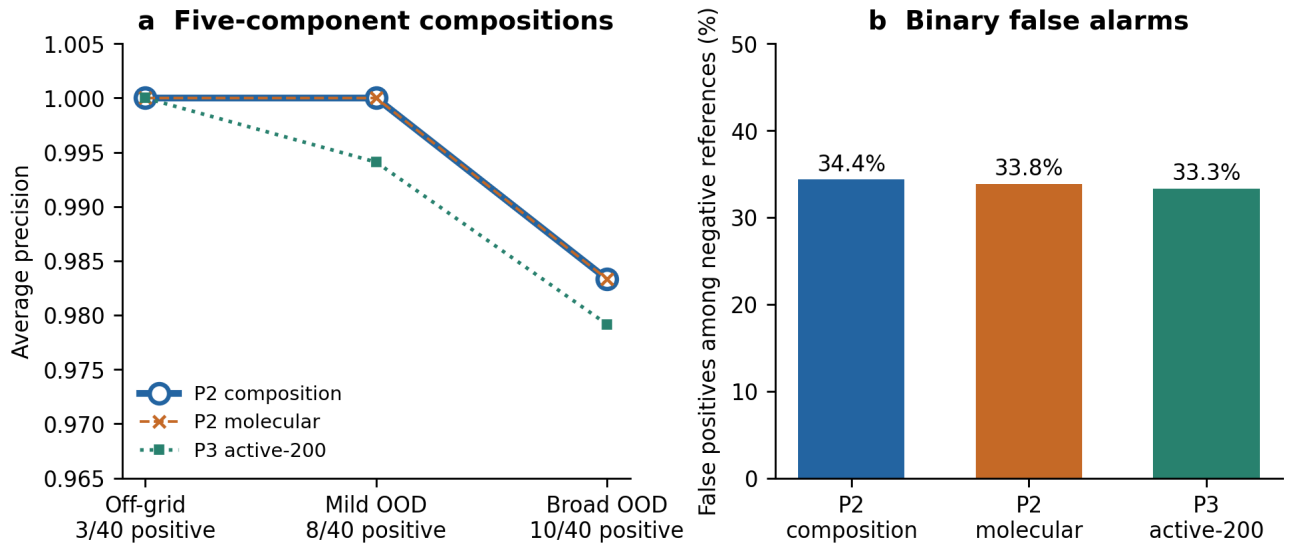


Figure 4. Useful five-component extrapolation coexisted with binary errors. Panel a shows AP in the three 40-recipe groups. P2 composition and P2 molecular have identical AP values in all three groups; blue open circles and solid lines are overlaid with orange crosses and dashed lines to show both series. Their recall and F1 can still differ (Table 5). Panel b shows false-positive rates among the 195 negative binary mixture references: 34.36% for P2 composition, 33.85% for P2 molecular, and a five-seed mean of 33.33% for P3 active-200.

These confident boundary errors motivated target adaptation and supported retaining random exploration in active acquisition. When committee members are confidently wrong in the same regime, their agreement provides little uncertainty signal to trigger a query. Random exploratory queries can reach such regions even when uncertainty-based selection assigns them low priority. This consideration supported retaining the random component alongside uncertainty and diversity in the later active-learning strategy.

The source models had learned the five-component interior region, with each solvent at 15–40 mol%, very well and remained useful on the broader five-component compositions tested. Their errors on binary faces and pure endpoints showed that this learning had not recovered reliable boundary behavior across the full composition simplex.

7. Adapting to the target domain with active selection and transfer

We next constructed a separate target dataset spanning binary, ternary, quaternary, and broader five-component compositions, with 600 training-pool, 200 validation, and 200 test recipes. The target test set contained 35 positives. Inspection of the frozen target recipes confirmed that all 1,000 recipes, including the 300 five-component compositions, lay outside the original source region (Methods M7), so the observed adaptation gains were obtained beyond the original source bounds, rather than being partly attributable to five-component target recipes lying inside them. A common reference dataset served all learning arms.

We compared scratch and transferred initialization, each with random or active acquisition, at 60, 150, and 300 target labels. A further scratch control received exactly the target compositions selected by transfer-active learning. This separated the benefit of initialization from the benefit of choosing different examples. A full-pool scratch model used all 600 target-training labels. Zero-shot and threshold-only evaluations measured how much could be achieved without changing the learned weights.

Table 6. Target adaptation performance. Values are five-seed means on the shared target test set.

Learning arm	Labels	AP	Recall	F1
Scratch random	150	0.8276	0.6400	0.7126
Transfer random	150	0.8432	0.5771	0.6876
Scratch active	150	0.9343	0.8743	0.8339
Scratch on transfer-selected examples	150	0.9352	0.7829	0.7967
Transfer active	150	0.9534	0.9314	0.8494
Scratch on transfer-selected examples	300	0.9398	0.9143	0.8145
Transfer active	300	0.9613	0.9486	0.8833
Full-pool scratch	600	0.9463	0.9200	0.8110

Transfer-active learning reached AP 0.9534 at 150 labels and 0.9613 at 300. Relative to scratch models trained on the same transfer-selected examples, its mean AP advantages were 0.0182 and 0.0215, positive in all five paired seeds at both budgets. F1 was higher in four of five paired seeds at 150 and all five at 300: nine of the ten seed–budget comparisons. These matched-example controls support a contribution from transferred initialization beyond the selected composition list. Separately, at 150 labels, scratch-active exceeded scratch-random by 0.1066 AP, showing the benefit of active selection. Together, the comparisons support roles for both transfer learning and active learning in improving target-domain performance.

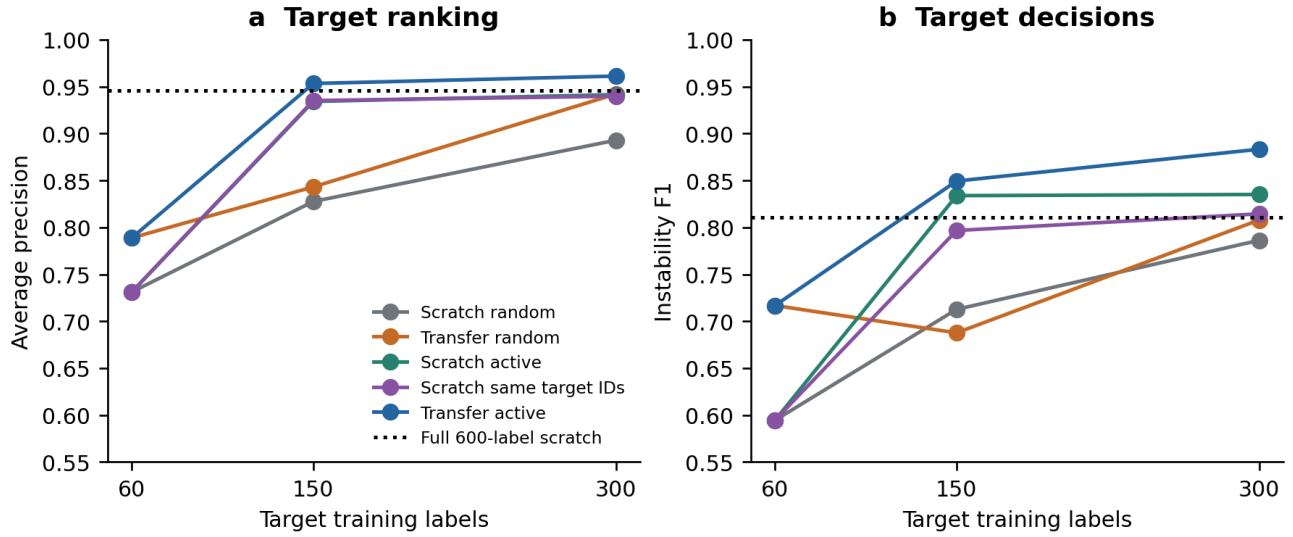


Figure 5. Target AP and F1 as functions of acquired target-training labels. Curves show five-seed means. The dotted line is the matched full-pool scratch reference. At the initial 60 labels, random and active arms within each initialization share the same starting data; their curves consequently coincide there.

The frozen P3 active-200 source committees had mean target AP 0.2903, compared with a positive prevalence of 0.175 (35 of 200 recipes). This slightly exceeded the constant-score prevalence baseline and suggested some retained ranking information, but the ordering was still weak. We evaluated both the original source cutoff and a cutoff selected using target-validation labels while holding the weights fixed. Threshold adjustment changed which recipes were flagged and therefore could change recall and F1, but it left the scores, their ordering, and AP unchanged. The target-domain difficulty thus extended beyond the choice of operating point. Within the tested workflow, learning from target labels changed the model weights and repaired the ranking, with transfer-active AP rising to 0.9534 and 0.9613 at 150 and 300 labels.

Target learning also corrected all eight sampled water–benzene test recipes, four positive and four negative, for every transfer-active seed at 150 and 300 labels. Scratch-active also classified those eight correctly. Both target-trained approaches therefore corrected these cases.

Domain-level results further clarified the tradeoff. At 150 labels, scratch-active had no binary false positives, whereas transfer-active averaged 3.6, but transfer-active had higher ternary recall, 0.80 versus 0.50. At 300 labels, transfer-active binary false positives decreased to 0.2 and ternary recall reached approximately 0.833. Aggregate AP therefore summarized different improvements across domains rather than uniform superiority on every subset.

The good target results concealed substantial source forgetting (also known as catastrophic forgetting). For the P3 active-200 sources, mean source AP decreased from 0.9930 before adaptation to 0.6698 at 150 target labels and 0.5937 at 300, which, as a ranking loss, could not be repaired solely by changing the source cutoff. This motivated testing whether a small memory of source examples could reduce forgetting (see Sections 9–11).

8. Interpreting the resource savings

Active-200 used 83.2% fewer source-training labels than the 1,191-label reference. Including the shared 314 validation labels changes the comparison to 514 versus 1,505, or a 65.8% reduction. Including the shared 495 test labels changes it to 1,009 versus 2,000, or 49.6%. For target transfer-active-150, the corresponding reductions were 75.0% for training labels, 56.25% including target validation, and 45.0% including target evaluation (Figure 6). These reductions count target labels; the transfer model also inherits source pretraining and its associated costs.

However, this distinction is particularly relevant when a pretrained source model already exists: a formulation program may accumulate a model of multicomponent mixtures (or more generally, items) through earlier empirical or computational work, then need to specialize it to binary or ternary mixtures (items), boundary compositions, or other restricted regimes. In such a top-down setting, transfer-active learning can reuse existing knowledge instead of rebuilding each target problem independently. Source development remains part of the total historical investment, while its already-incurred cost is separate from the incremental cost of the new adaptation task.

Table 7. Label and recorded-duration accounting. Durations are sums of recorded per-recipe elapsed times, averaged across selection seeds where applicable. They are neither campaign wall times nor measured CPU-hour totals.

Comparison	Selected strategy	Full-pool comparator	Reduction
P3 source-training labels	200	1,191	83.2%
P3 training plus validation labels	514	1,505	65.8%
P3 labels including evaluation	1,009	2,000	49.6%
P3 training reference-duration sum	5.130 h	27.775 h	81.5%
P4 target-training labels	150	600	75.0%
P4 target training plus validation labels	350	800	56.25%
P4 target labels including evaluation	550	1,000	45.0%
P4 training reference-duration sum	6.370 h	23.024 h	72.3%
P4 training plus validation duration sum	13.501 h	30.155 h	55.2%

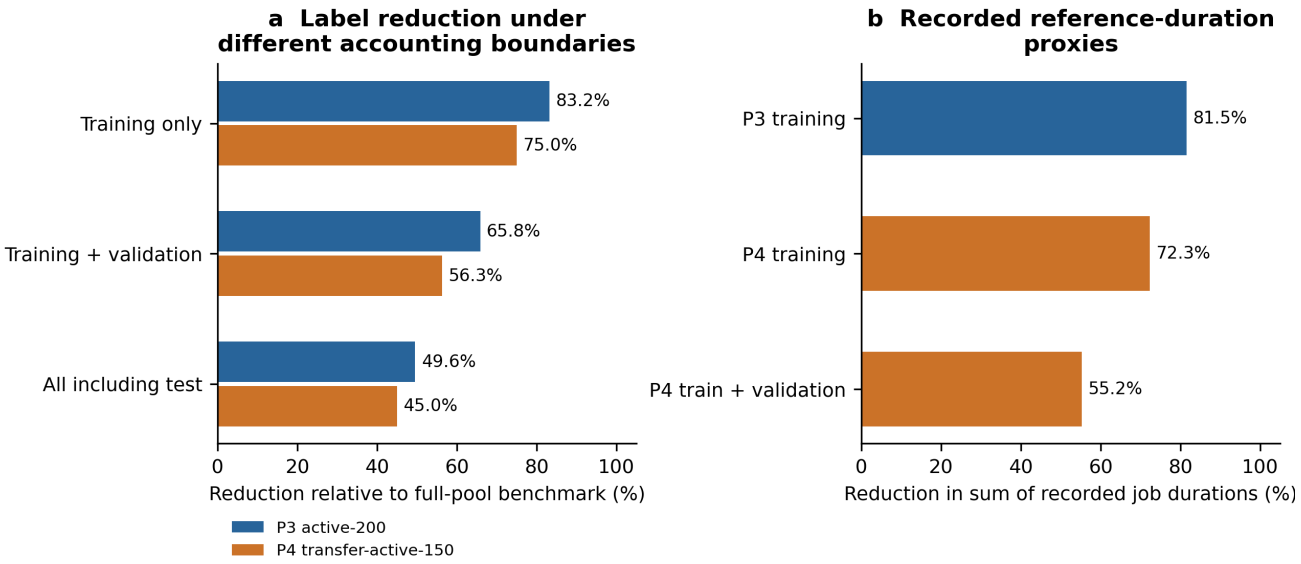


Figure 6. Resource savings under different accounting boundaries. Panel a shows reductions in required reference-label counts, with training alone, training plus validation, and training plus validation and evaluation reported separately. Panel b compares summed historical reference-search durations with full-pool benchmarks, including validation where indicated. Duration reductions are slightly smaller than label-count reductions under matching accounting boundaries because the selected recipes took longer to evaluate on average. These are not end-to-end wall-clock speedups. Cumulative fitting took seconds to minutes, compared with hours of summed reference-search durations; transfer-active fitting nevertheless took longer than the single full-pool scratch fit.

9. Retention, Stage 1, asking whether a larger source model is sufficient

Before selecting replay memories, we examined whether the source model itself changed the forgetting problem. We compared P3 active-200, P2 composition, and P2 molecular after target-only fine-tuning, then measured their retention on the original source test set (Methods M7–M8). Within each seed and budget, all three source families were adapted using the same target-training examples from the saved transfer-active acquisition path, without source replay or new acquisition. The completed P3 target-adapted models from Section 7 were reused; the P2 source committees were fine-tuned on those same subsets. Each fit started from the corresponding original source checkpoint and updated the network weights. P2 used 1,191 source-training examples, while each P3 source used its own 200 acquired examples. P2 molecular also differed in representation and original class-weighting history. These comparisons concern practical saved source models and do not isolate source-training size alone.

Without replay, the larger molecular source still forgot substantially. Its source AP was 0.6164 at 150 target labels and 0.6463 at 300. P2 composition retained more in these comparisons, with source AP 0.8310 and 0.7392 and target AP 0.9810 and 0.9825. Thus, a larger source set and the simple molecular descriptors did not by themselves prevent substantial source forgetting.

For subsequent replay experiments (Sections 10-11), every feasible policy retained 20 source examples and reused that memory at target budgets 60, 150, and 300. Target examples were identical across replay policies within each seed and budget. Replay received 20% of the total supplied sample weight, with the remaining 80% allocated to target examples. Only source-training recipes correctly classified by the frozen source teacher were eligible; P3 eligibility was restricted to that seed's acquired 200 examples. Methods M8–M9 specify selection, weighting, preprocessing, and the distinction between repeated adaptation of one P2 source committee and adaptation of five separately acquired P3 sources.

The eligible source memories revealed a common association underlying the two guided strategies. Among 20 selected examples, Slowest-20 contained 14 positives for P2 molecular and a mean of 12.2 for P3 active-200; Highest-energy-20 contained 20 and 17.4, respectively, compared with 1.8 and 4.0 for uniform random replay. Slowest-20 also had higher mean predicted mixing energy than the random memories, while Highest-energy-20 had longer summed recorded search durations. Thus search duration, energy ordering, and positive-label enrichment were associated in these selected subsets. This supports reusing the computational history alongside thermodynamic predictions to guide memory selection; the class-matched controls in Sections 10–11 test whether the particular examples add value beyond enrichment alone.

10. Retention, Stage 2, using search duration to choose reminders

Our first guided-memory hypothesis was that examples with long reference-search times might be particularly valuable to preserve. A slow calculation may reflect a numerically difficult search region, although execution conditions also influence recorded durations. Slowest-20 therefore used duration as an empirical selection signal, retaining the 20 eligible source recipes with the longest recorded stability searches. Uniform Random-20 drew from the same eligible pool, while a time class-matched random control reproduced the positive and negative counts selected by Slowest-20. All memories reused previously labeled examples.

The P2 Slowest-20 memory contained 14 positives and six negatives, compared with a mean of 1.8 positives among the 20 uniformly sampled examples. For P3, the corresponding mean positive counts were 12.2 and 4.0. P3 had a higher positive fraction in its acquired source-training sets (19.5% on average, versus 7.6% for P2, before teacher-correct filtering; Section 11), which together with the smaller and actively acquired P3 pools, limits direct comparisons between the two source families, although both are informative settings for evaluating duration-guided and class-matched replay. The class-matched strategy inherited the positive enrichment induced by duration ranking while randomizing example identities. The two strategies form a related family of duration-associated memory constructions. Comparing Slowest-20 with uniform random replay examines the combined effects of class allocation and example selection; comparing its class-matched control with uniform random replay examines the value of the induced class allocation under random example selection; and comparing

Slowest-20 with its matched control examines the additional value of selecting the particular longest-running examples.

For P2 molecular, one pretrained source committee and the same Slowest-20 memory were reused across adaptation seeds 11, 22, 33, 44, and 55; the random memories varied by seed. Thus the five P2 runs assess a fixed memory under different adaptation runs, rather than five independently selected source pools. For P3 active-200, each seed had its own pretrained committee, acquired source subset, and selected memory. Within each seed and target budget, the target-training examples were identical across replay policies (Methods M8).

Table 8. Duration-guided replay and its controls at all three target-label budgets. Values are means across five adaptation seeds with transferred initialization. AP is average precision; recall and F1 refer to the positive instability class. Each fitted model's target-validation cutoff is used on both separate test sets (source: 495 recipes, 33 positives; target: 200 recipes, 35 positives). "Matched" denotes time class-matched replay. The seed design is described above and in Methods M8.

Target labels	Memory	Source AP	Source recall	Source F1	Target AP	Target recall	Target F1
P2 molecular							
60	No replay	0.7858	0.4606	0.4525	0.7589	0.6000	0.5763
60	Random-20	0.8138	0.3273	0.3265	0.7683	0.5371	0.5684
60	Slowest-20	0.7899	0.5879	0.5782	0.7492	0.5657	0.6197
60	Matched	0.8772	0.8485	0.5879	0.8427	0.7543	0.7135
150	No replay	0.6164	0.7333	0.4310	0.9471	0.8343	0.7773
150	Random-20	0.7637	0.6848	0.6057	0.9154	0.7371	0.8019
150	Slowest-20	0.8601	0.5758	0.6545	0.8730	0.6686	0.7414
150	Matched	0.8112	0.8303	0.5637	0.9518	0.7657	0.8251
300	No replay	0.6463	0.8727	0.4785	0.9578	0.9143	0.8512
300	Random-20	0.8204	0.7939	0.6874	0.9390	0.8171	0.8597
300	Slowest-20	0.8693	0.7333	0.7348	0.9089	0.8571	0.8515
300	Matched	0.8318	0.9879	0.5178	0.9525	0.9371	0.8377
P3 active-200							
60	No replay	0.9094	0.5091	0.3659	0.7888	0.7486	0.7167
60	Random-20	0.9435	0.6727	0.7660	0.7415	0.6743	0.6466
60	Slowest-20	0.9844	0.6606	0.6810	0.7599	0.6743	0.6273
60	Matched	0.9463	0.7697	0.6264	0.7921	0.8000	0.7084
150	No replay	0.6698	1.0000	0.4210	0.9534	0.9314	0.8494
150	Random-20	0.9481	0.9515	0.8624	0.9135	0.8571	0.8258
150	Slowest-20	0.9628	0.9091	0.8517	0.9126	0.8571	0.8958
150	Matched	0.9431	0.9636	0.7249	0.9167	0.8571	0.8174
300	No replay	0.5937	0.9939	0.3262	0.9613	0.9486	0.8833
300	Random-20	0.9529	0.9455	0.8502	0.9214	0.8686	0.8423
300	Slowest-20	0.9690	0.8909	0.8663	0.9136	0.8229	0.8647
300	Matched	0.9509	0.9697	0.7458	0.9306	0.8629	0.8574

Considering the related policies together reveals a favorable mean source-retention pattern. The equal-weight average of Slowest-20 and its class-matched control exceeded uniform Random-20 in source AP in all six source-family/budget combinations. This average summarizes the two evaluated policies. In P3, the within-seed two-policy average exceeded Random-20 in only two of five seeds at each budget. On the target test set, class-matched

replay exceeded uniform random replay in mean AP across all six comparisons, although the two-policy average did not consistently outperform random replay.

For P2 molecular, Slowest-20 achieved higher mean source AP than both Random-20 and class-matched replay at 150 and 300 target labels, though lower mean source AP at 60 labels, despite outperforming Random-20 in three out of five seeds; its mean target AP remained below both controls at all three budgets. For P3 active-200, Slowest-20 exceeded both controls in mean source AP at all three budgets, with three of five paired seeds favoring it over Random-20 at each budget. Despite favorable mean source-AP differences, Slowest-20 outperformed its class-matched control in only two out of five paired seeds at 150 and 300 target labels in either source family; nevertheless, P2 Slowest-20 achieved higher source F1 than its class-matched control in four out of five paired seeds at both 150 and 300 labels. For P3 at 150 labels, Slowest-20 also improved target F1 in all five seeds relative to its class-matched control, despite lower target AP than class-matched, Random-20, and no replay. At 60 labels, all five P3 seeds favored class-matched replay over Slowest-20 in target F1.

The practical finding is that recorded search duration can inform replay through example selection (Slowest-20 replay) and through the resulting class allocation revealed by that selection (class-matched replay). Their relative usefulness depended on the source pool, budget, and metric; both approaches demonstrated the potential value of reusing information generated during thermodynamic instability screening.

11. Retention, Stage 3, selecting an energy bridge

The second hypothesis used predicted molar mixing free energy to select inputs to the replay memory. We applied the frozen Project 1 Ridge model to the eligible source compositions, restored the analytical ideal contribution, and sorted by total predicted mixing free energy from highest (least negative) to lowest.

We tested the 20 highest-energy examples, the 20 lowest-energy examples, and class-matched random controls. At 300 target labels, P3 highest-energy replay preserved source ranking well (AP 0.9419), unlike lowest-energy replay (0.6244). For P2 molecular, neither extreme retained source ranking well, although highest-energy replay achieved strong target AP (0.9783). Interestingly, randomly selected negative-only memories preserved source ranking substantially better than Lowest-20 in both families (P3: 0.9409 versus 0.6244; P2: 0.8066 versus 0.6719 at 300 labels), with all five paired seeds favoring the random-negative control. For highest-energy replay, a separate random control matched its positive and negative counts while sampling from the full eligible source pool. P3 active-200 highest-energy replay achieved higher mean source AP than this control at 150 labels (0.9505 versus 0.9073) and 300 labels (0.9419 versus 0.8692), with three of five paired seeds favoring it at each budget. In P2 molecular, both highest-energy replay and its matched control retained source ranking poorly at these budgets. These patterns provide a rationale for examining a memory spanning both labels near the higher-energy transition.

The bridge tested a more focused hypothesis: retain contrasting labels around the first transition in the descending-energy list. The strict memory contained ten positives immediately before the first negative and ten negatives encountered from that point downward, skipping intervening positives; a random memory containing ten positives and ten negatives from the same eligible pool controlled for class composition (Methods M8).

In the larger P2 molecular pool, the strict bridge produced the clearest energy-specific retention result. At 150 and 300 target labels, source AP was 0.9518 and 0.9452, exceeding the balanced control by 0.1612 and 0.1415, respectively. Four of five paired adaptation seeds favored the bridge over the balanced control at each budget.

Table 9. The P2 strict bridge and balanced control. Each result summarizes five adaptation seeds. AP differences are bridge minus control.

Target labels	Bridge source AP	Control source AP	Source AP difference	Bridge target AP	Control target AP	Target AP difference
60	0.8296	0.8208	+0.0088	0.8395	0.8147	+0.0247
150	0.9518	0.7906	+0.1612	0.9328	0.9612	−0.0284
300	0.9452	0.8037	+0.1415	0.9312	0.9535	−0.0223

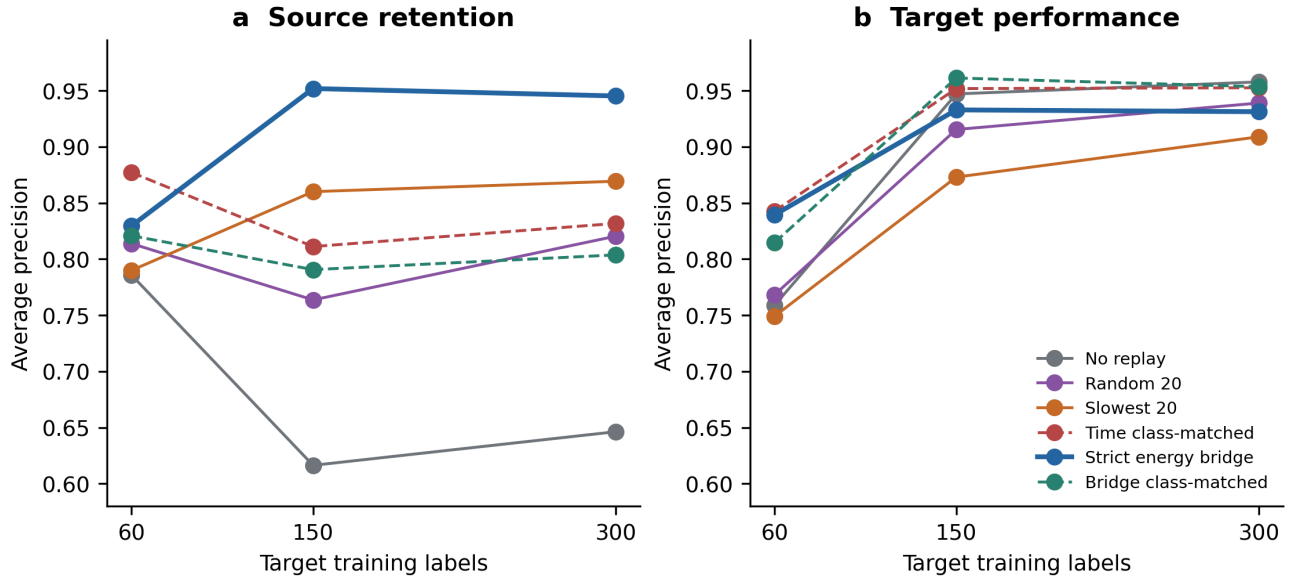


Figure 7. P2 molecular replay across target budgets. The strict bridge gives substantially stronger mean source retention than its balanced control at 150 and 300 labels. Both domains are shown because some of that retention comes with reduced target AP. Lines are five-seed means; Figure 8 shows the paired bridge effects.

For P2 molecular, the strict bridge exceeded Slowest-20 in both mean source and mean target AP at 150 and 300 labels. At 300 labels, pretrained initialization achieved mean source/target AP of 0.9452/0.9312, compared with 0.8938/0.8867 for scratch initialization using the same bridge, target examples, and source-fitted preprocessing. This supports a benefit from transferring the pretrained weights under these shared controls; the scratch comparison still inherited the source-guided memory and preprocessing. Together, these results support retaining contrasting labels near the energy-ranked transition for P2 source retention at the

two larger target budgets, beyond selecting energy extremes or random examples with the same 10/10 class balance.

The strict rule, requiring ten positives immediately before the first negative in descending energy order, was feasible for only one P3 active-200 seed, 55. At 150 and 300 target labels, the bridge increased source AP from 0.6995 to 0.9689 and from 0.6446 to 0.9754 relative to no replay, respectively. These single-seed results in active-200 demonstrate substantial retention, with mixed evidence for an additional benefit from energy-organized selection relative to the balanced control.

Table 10. Strict P3 bridge and controls for the single feasible source seed, 55. AP, instability recall, and instability F1 are evaluated on the same source and target test sets using each fitted model's target-validation cutoff.

Target labels	Replay policy	Source AP	Source recall	Source F1	Target AP	Target recall	Target F1
60	None	0.9004	0.0000	0.0000	0.8088	0.6286	0.7586
60	Strict bridge	0.9925	0.5455	0.7059	0.7966	0.6571	0.7667
60	Balanced control	0.9863	0.7576	0.8621	0.7736	0.6571	0.7419
150	None	0.6995	1.0000	0.5841	0.9755	0.9429	0.9565
150	Strict bridge	0.9689	0.8788	0.9062	0.9349	0.8286	0.9062
150	Balanced control	0.9972	0.9394	0.9538	0.9502	0.8857	0.9394
300	None	0.6446	1.0000	0.3837	0.9823	0.9714	0.9189
300	Strict bridge	0.9754	0.9091	0.8955	0.9515	0.9143	0.8533
300	Balanced control	0.9972	1.0000	0.8800	0.9565	0.9143	0.8312

We created an energy-ordered 20-example memory for every P3 seed by taking the first ten positives encountered from the high-energy end, skipping interspersed negatives, then the first ten negatives strictly after the tenth selected positive (Methods M8). A balanced random memory served as the control. At both 150 and 300 labels, the extension improved source AP over no replay in all five seeds, while all five lost target AP. Mean source AP was 0.9762 and 0.9763, versus 0.6698 and 0.5937 without replay. Relative to the balanced control, mean source AP was higher at 60 labels but lower at 150 and 300 (Table 11). Mean source F1 was higher at all three budgets using the target-validation cutoffs. Compared with Random-20, the downstream bridge improved mean source AP at every budget (0.9811/0.9762/0.9763 versus 0.9435/0.9481/0.9529 at 60/150/300 labels), with higher mean target F1 at all three budgets and source F1 at two. These results support the downstream bridge as a useful alternative to unconstrained random replay, while its advantages over class-matched selection depended on the metric and budget.

Table 11. P3 downstream extension across all five seeds. Both policies use ten positives and ten negatives. Differences are downstream bridge minus balanced control.

Target labels	Bridge source AP	Control source AP	Source difference	Bridge target AP	Control target AP	Target difference
60	0.9811	0.9558	+0.0253	0.7671	0.7865	−0.0194
150	0.9762	0.9831	−0.0070	0.9215	0.9163	+0.0052
300	0.9763	0.9834	−0.0071	0.9206	0.9285	−0.0079

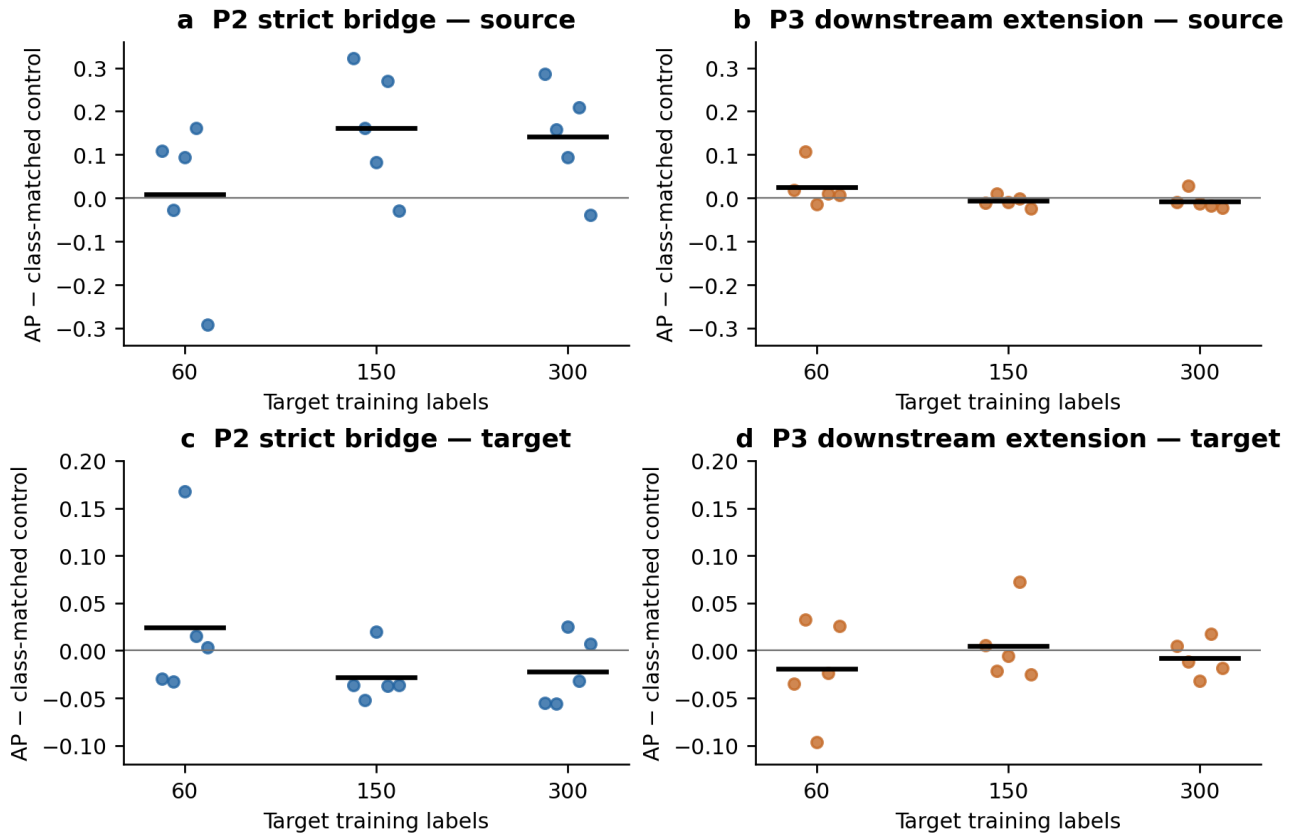


Figure 8. Paired AP changes relative to the respective balanced control. Each point is one adaptation seed; the black horizontal segment is the mean. The left column shows the P2 strict bridge and the right column the five-seed P3 downstream extension. Source and target have different vertical scales. The one-seed strict P3 experiment is discussed separately and is not combined with the five-seed extension.

Several differences between the source pools may help explain the stronger energy-specific retention result in P2 molecular. P2 used 1,191 source-training examples, including 91 positives (7.6%), whereas P3 active-200 used 200 examples per seed, averaging 39 positives (19.5%; range 35–47), before teacher-correct filtering (i.e., choosing examples that the pretrained model classified correctly according to its known reference labels). Although P3 had a higher positive fraction, its smaller, actively acquired pools contained fewer positives before the first negative in the energy ranking, limiting strict-bridge feasibility. Pool size and acquisition history may therefore have influenced the relative effectiveness of energy-guided and class-balanced random replay. Representation and pretraining also differed between P2

and P3, and the extension changed the bridge construction. Future investigations could examine adaptive bridge allocations or include interspersed negatives in the downstream selection, with corresponding class-matched controls.

12. Conclusions, connecting the findings, and defining the remaining questions

The sequence provides several linked findings. First, a simple composition-based polynomial captured the calculated excess-energy surface extremely well in the source region and remained useful beyond its original bounds. Second, instability classification was possible with inexpensive features, while feed activity coefficients improved ranking without requiring the full stability search. Third, active acquisition substantially reduced the number of source labels needed to reach strong performance. Fourth, the frozen classifiers generalized well to previously unseen five-component compositions beyond the original source bounds but made confident errors near binary limits, motivating targeted adaptation. Finally, target adaptation improved performance in the new composition region but degraded source performance, motivating the replay experiments.

Search duration informed replay construction (Section 10). The exploratory equal-weight mean source AP of Slowest-20 and its class-matched control exceeded uniform random replay in all six source-family/budget combinations. These results support considering both search duration and the induced class allocation, with benefits dependent on the setting and metric.

The P2 strict bridge (replay into the P2 molecular classifier, with energies supplied by the frozen Ridge surrogate) gave the energy model a second role. The surrogate first enabled inexpensive energy evaluation; later, its predictions organized a small memory that preserved mean source ranking better than a balanced random control. The observed gains support an energy-specific retention benefit in this setting, together with a measurable target-AP tradeoff. P3 active-200 gave a more mixed but still useful result in the energy bridge setting. Its downstream bridge preserved source ranking better on average than ordinary Random-20 at every tested budget, with higher target F1, but did not consistently exceed the balanced control. The single feasible strict-bridge seed also showed substantial retention relative to no replay and higher source and target F1 than its balanced control at 300 labels. Separately, random-negative memories outperformed the lowest-energy negative-only memories in source AP in all five seeds of both families at 300 labels (Section 11).

The broader practical opportunity of our approach is to map a chosen composition region without running an expensive instability search at every candidate. In the finite benchmarks, active-200 used 83.2% fewer source-training labels than the full 1,191-example pool, and transfer-active-150 used 75.0% fewer target-training labels than the 600-example target pool (Section 8). Once trained and evaluated for the intended regions, inexpensive predictors can screen additional compositions and direct reference calculations toward selected candidates. Because instability searches can be considerably more expensive than surrogate inference,

this approach offers a practical route to computational thermodynamic screening while offering an interesting testbed for machine-learning methods.

A natural later extension would acquire up to 2,000 training labels through active learning from an initial candidate pool of approximately 20,000 compositions, in order to test strict energy bridges in larger acquired source pools. Under matched representations and memory budgets, it could in addition compare strict, downstream, and alternative bridge constructions, balanced controls, and the untested lower-energy bridge. Increasing the actively acquired source-pool size while keeping the candidate pool, model representation, and memory size fixed would help establish whether strict energy bridges become more feasible and effective with additional source examples. Extending these results to new molecular identities, temperatures, and experimentally measured phase or instability behavior could be an important next step.

The energy-bridge result suggests a broader hypothesis for imbalanced classification under distribution shift: a small replay memory may benefit from retaining contrasting labeled examples near transitions in an inexpensive, domain-informed ordering. Beyond molecular mixtures, candidate orderings could arise from interaction costs or surrogates of expensive optimization objectives. Such constructions are particularly relevant when evaluating an initial or a reference configuration is cheaper than searching over competing configurations to determine its stability label. The present study provides evidence for this strategy in thermodynamic source retention.

Methods

M1. Molecular preparation and thermodynamic reference

We considered water, methanol, acetone, acetonitrile, and benzene in that fixed order. Mole fractions sum to one; an absent component has zero fraction and is excluded from the corresponding reference phase calculation. All reported mixture comparisons refer to 298.15 K. Preliminary water–methanol calculations used published .orcacosmo reference surfaces from the openCOSMO-RS 24a dataset [2,13]; the quantitative campaigns reported here used our independently calculated molecular surfaces.

Starting geometries were generated with RDKit ETKDGv3 (seed 42) and optimized with MMFF94. ORCA 6.1.1 [14], driven through the ORCA Python Interface [15], then performed BP86/def2-TZVP geometry optimization with TightSCF and TightOpt in a conductor environment. The subsequent COSMORS(WATER) wrapper used BP86/def2-TZVPD and selected water as the auxiliary solvent for ORCA's bundled solvation calculation [9]. We retained each molecule's conductor-phase .solute.orcacosmo surface and did not use the printed water-solvation free energy. These surfaces were subsequently combined in openCOSMO-RS 24a [2] at explicitly specified multicomponent mole fractions. Water was therefore one component of the later mixtures (without a distinguished solvent role). We used one starting geometry per molecule, without an exhaustive conformer search or frequency-based certification of minima. Surface files and their recorded hashes define the molecular reference used by the numerical campaigns.

Activity coefficients were calculated relative to pure-liquid standard states. Using consistent energy units, the molar quantities were

$$g^E(x,T)=RT\sum_i x_i \ln \gamma_i(x,T), g_{id}(x,T)=RT\sum_i x_i \ln x_i,$$

$$g_{mix}=g^E+g_{id}.$$

We use the limiting convention $x\ln x=0$ at $x=0$. At fixed composition, excess entropy was estimated by a central temperature difference,

$$s^E(T)\simeq -\frac{g^E(T+\Delta T)-g^E(T-\Delta T)}{2\Delta T}, h_{mix}=g^E+T s^E,$$

$$s_{mix}=s^E-R\sum_i x_i \ln x_i.$$

The primary step was $\Delta T=1$ K, checked against 2 K with unchanged molecular surfaces. Thus one derivative stencil uses three temperatures, and the combined check uses five distinct temperatures: 296.15, 297.15, 298.15, 299.15, and 300.15 K. For the production reference calculations, we explicitly set the COSMOSPACE convergence tolerance to 10^{-10} with an

iteration limit of 3,000. Energies were stored in kJ/mol and entropy in J/(mol K), with the corresponding unit conversion in the enthalpy calculation.

M2. Composition descriptors and source datasets

Four independent mole fractions were supplied to the composition-only learners; the fifth, benzene, was reconstructed as $x_5 = 1 - \sum_{i=1}^{4} x_i$. Molecular descriptors were deterministic functions of the five fractions and fixed component constants. They comprised mean molar mass, mean total atom count, four elemental atom counts (H, C, N, O), five component-specific atom contributions, additive molar volume, and additive density. For component molar masses M_i and pure-liquid density constants ρ_i,

$$V_{add} = \sum_i x_i \frac{M_i}{\rho_i}, \rho_{add} = \frac{\sum_i x_i M_i}{V_{add}}.$$

The density constants, in solvent order, were 0.9970, 0.7866, 0.7845, 0.7767, and 0.8736 g/mL. They approximate pure-liquid values at 298.15 K near atmospheric pressure. Atomic masses were H 1.008, C 12.011, N 14.007, and O 15.999 g/mol. This volume-additive construction neglects excess volume.

For Project 1, the source domain contained every 1 mol% grid composition with each fraction between 0.15 and 0.40, giving 23,751 recipes. Blocks were defined by 5 mol%-wide bins of the first four fractions: $b_i = \lfloor (100\, x_i - 15)/5 \rfloor$. A SHA-256 hash of the campaign seed and block identifier assigned whole blocks to training, validation, or test using nominal 70/15/15 hash buckets. With seed 88, the realized sizes were 17,121, 2,872, and 3,758. Grouping neighboring recipes prevents the split from being a random scattering of individual grid points.

For Project 2, the initial screen included all 126 coarse 5 mol% grid recipes and 474 sampled fine-grid recipes. A further 1,400 recipes expanded the labeled set to 2,000. Only accepted positive and negative search outcomes were eligible for classification; no unresolved cases were reported in this 2,000-recipe campaign. The final classification split was constructed separately from the regression split, using the same composition-block concept. Four-fold stratified grouped splitting with seed 2026 reserved the first held-out fold as test. Among five grouped folds of the remaining development set, validation was chosen by closeness to 20% of each development class, using class counts rather than model scores. The resulting training, validation, and test sets contained 1,191/314/495 recipes and 91/30/33 positives. Three grouped training-only folds, generated with seed 2027, were then fixed for model selection.

M3. Reference instability labels

We searched for an instability of each homogeneous feed using the tangent-plane criterion [4]. For feed composition z and trial-liquid composition w, the dimensionless objective was

$$D(w;z)=\sum_i w_i\left[\ln\{w_i\gamma_i(w)\}-\ln\{z_i\gamma_i(z)\}\right].$$

The activity model and temperature were shared by the feed and trial liquid. SLSQP minimized this objective subject to the sum-to-one constraint and component bounds, with a numerical floor of 10^{-10} [12]. Trial compositions could leave the bounded source-feed region. The implementation normalizes the optimizer variables and supplies the corresponding analytical gradient using the Gibbs–Duhem relation. Its stopping tolerance was 10^{-10} and its iteration limit was 300 per start.

The reported labeling campaigns used 40 starts per feed. Starts included the feed, one component-rich composition per present component, one pair-rich composition per component pair, and random Dirichlet compositions to fill the remaining starts. Component-rich starts used $0.95\,e_i+0.05/n$; pair-rich starts assigned an additional 0.475 to each selected component on top of $0.05/n$. Random starts cycled through symmetric Dirichlet concentrations 0.25, 1, and 3. The lowest objective encountered was retained even if an optimizer subsequently failed to converge. Search seeds and realized recipes are part of the campaign records.

A candidate with $D<-10^{-6}$ was checked by forming a small, mass-balanced second liquid. For trial-phase fraction β,

$$z=\beta w+(1-\beta)v,\ v=\frac{z-\beta w}{1-\beta},$$

$$\Delta g=\beta g_{mix}(w)+(1-\beta)g_{mix}(v)-g_{mix}(z).$$

The initial fraction was $\beta=min\left[0.01,0.5\,min_i(z_i/w_i)\right]$ over present components. Up to ten checks reduced this fraction by a factor of ten when needed. An accepted positive required a negative tangent-plane candidate and a verified $\Delta g<-10^{-10}$ kJ/mol. An accepted negative required no candidate below the tangent-plane tolerance and convergence of every local search. Other cases required review. The positive label therefore means a verified energy-lowering split was detected; the negative label means none was detected by this finite search. Neither the trial phases nor a negative label constitute a complete equilibrium phase diagram or a global stability proof. Pure endpoints were treated as analytic negative checks.

M4. Energy regression and frozen extrapolation

Project 1 predicted g^E from the four independent fractions. We compared a training-mean baseline, polynomial Ridge regression, and histogram gradient boosting. Polynomial features excluded a constant bias feature and were standardized using the training subset before

fitting Ridge. Boosting used the unexpanded fractions. Hyperparameters were selected by mean validation MAE within each family and training size; validation was not absorbed into the final training set. The learning curves used 100, 300, 1,000, 3,000, 10,000, and all 17,121 training recipes, with subset seeds 42, 88, and 123. The final full-data choices were frozen before test evaluation. Appendix A gives the search grid and selected settings.

Total mixing Gibbs energy was reconstructed by adding the exact ideal contribution to the predicted excess energy. Consequently, the excess- and total-energy absolute errors coincide recipe by recipe. MAE, RMSE, and R^2 describe agreement with the openCOSMO-RS reference. No separate enthalpy or entropy predictor was trained in this experiment.

Frozen extrapolation used 200 new off-grid recipes within the original bounds, 200 mild-OOD recipes within 0.10–0.45 but outside the source bounds, and 200 broad-OOD recipes within 0.01–0.90 but outside the mild region. A symmetric Dirichlet draw was shifted by the lower bound and rejected if it violated the upper bound or belonged to an excluded region. Duplicate compositions were removed. The water–benzene challenge used 101 fractions from zero to one in steps of 0.01, supplemented by five trace fractions (10^{-6}, 10^{-5}, 10^{-4}, 0.001, 0.005) and their complements, giving 111 points including pure endpoints. Two named five-component examples were also retained in the archive; they were not pooled into the principal group comparisons in Table 2.

M5. Instability classifiers and evaluation rules

Project 2 compared a majority baseline, logistic regression, an RBF support-vector classifier, and MLP committees. Feature groups were composition (4 inputs), molecular (17), thermodynamics (7), activity (9), and full (25). The thermodynamic group added g^E, h_{mix}, and s_{mix} to composition; the activity group added the five feed $\ln \gamma_i$ values. Full features combined composition, the 13 molecular descriptors, three thermodynamic quantities, and five log activity coefficients. Optimized trial-phase properties, search minima, labels, identifiers, and search durations were excluded from prediction inputs.

Each fitted classifier pipeline standardized inputs using its own training fold or training set. Candidate settings included unweighted fits and inverse-frequency class-weighted fits; weights were calculated from the labels supplied to that fit. Hyperparameters maximized mean average precision across the three fixed grouped training folds. MLP candidates were assessed using initialization seeds 88 and 2026; the final selected MLP comprised three independently initialized members with seeds 88, 2026, and 31415. Their positive-class scores were averaged. The RBF classifier used its decision function without an additional internal probability-calibration split.

After training-only refitting, a decision cutoff maximized instability F1 on the fixed validation set, with ties favoring higher recall and then the lower cutoff. The majority baseline retained its majority decision. Selected settings and cutoffs were saved before test evaluation. Average precision measures ranking; instability recall and F1 use the frozen cutoff and treat detected

instability as the positive class. Where a subset contains no positives, positive-class AP and recall are not interpreted as meaningful performance estimates.

For multiseed experiments, we report seed means and, where shown, sample standard deviations and paired differences at the same budget and on the same evaluation recipes. P2's saved uncertainty procedure resampled whole test blocks 1,000 times; those intervals describe the fitted model on this benchmark. The source test set was reused across the linked projects, and later experiments were informed by earlier findings. Evaluation labels did not enter the specified acquisition, fitting, or cutoff-selection procedures within each frozen run.

M6. Source active learning

Project 3 was a retrospective pool-based experiment on the existing 1,191 source-training recipes. Only queried labels were revealed to the acquisition learner. Validation and test recipes were excluded from the pool. For seeds 11, 22, 33, 44, and 55, each policy started from the same label-blind random set of 100 recipes within that seed. Acquisition added batches of 25, with reporting checkpoints at 100, 200, 400, and 800. Random acquisition, Euclidean farthest-first space filling, and uncertainty plus diversity were compared. Geometry used all five fractions even though the MLP used four independent inputs.

For committee scores $p_m(x)$, we defined binary entropy H in bits and the acquisition score

$$u(x) = H\left(p_{avg}\right) - \frac{1}{M}\sum_{m=1}^{M} H\left(p_m\right) + 0.25\, H\left(p_{avg}\right), p_{avg} = \frac{1}{M}\sum_{m} p_m.$$

The first two terms measure committee disagreement through the Jensen–Shannon divergence between Bernoulli predictions; the last adds prediction ambiguity. For a batch of 25, the algorithm shortlisted the 100 highest-scoring unqueried candidates, selected 20 by farthest-first diversity relative to the revealed set and newly selected candidates, and added five uniformly random unqueried candidates. Randomized tie handling avoided dependence on table order. If scores were constant, the diversity stage operated on the remaining pool.

At each acquisition round, a new three-member MLP committee and training-only scalers were fitted on all revealed recipes. The prescribed architecture had hidden layers of 128 and 64 units, ReLU activation, Adam optimization, learning rate 0.001, and regularization parameter 0.01. Inverse-frequency weights used revealed labels only. A single-class revealed set used constant predictions until further acquisition. The code allowed one training-only continuation after an epoch-limit warning; it did not carry fitted weights between acquisition rounds. Appendix A summarizes the documented epoch and continuation settings.

A matched full-pool committee was fitted independently for each seed after acquisition paths had finished. Validation cutoffs were then fixed before test scoring. The first evaluated checkpoint satisfying AP at least 0.98 and recall at least 0.90 was reported as a label-efficiency milestone. This retrospective test-set summary did not control acquisition or stopping and should not be interpreted as an online guarantee.

M7. Frozen classification challenge and target adaptation

The frozen P4 classification challenge evaluated saved source models, preprocessing, and source cutoffs without refitting. Its reported 335 recipes comprised 40 off-grid source-region, 40 mild-OOD, and 40 broad-OOD five-component recipes, 21 recipes for each of ten binary pairs, and five pure endpoints. The archive additionally contains two named five-component examples; the reported 335-recipe subset excludes them. Thus 330 reported recipes required searches and five were analytic endpoint checks. This challenge preceded and informed the target-adaptation study.

The separate target dataset contained 1,000 recipes, split by composition blocks into 600 training, 200 validation, and 200 test recipes. It included 400 binary, 200 ternary, 100 quaternary, and 300 five-component recipes. Binary compositions were drawn within ten equal logit intervals spanning fractions from approximately 10^{-6} to $1-10^{-6}$, with four recipes per interval for each pair. Validation and test each received one inner and one outer interval per pair. For each ternary and quaternary subset, ten Dirichlet anchors defined local composition blocks with two accepted recipes per block; the five-component set used 50 anchors with six recipes per block. Anchor concentrations cycled through 0.25, 0.5, 1, and 2. Local draws used Dirichlet parameters $150\,a+0.05$, were shifted to a 10^{-6} floor on present components, and were accepted only within their anchor's nearest-anchor region. Whole blocks were assigned to splits without reference labels. Seed 20261008 defined the target generator.

The frozen target recipe file confirms that all 300 five-component compositions lay outside the source bounds. This was a property of the realized sample, not an explicit rejection condition in the target generator.

Target learning compared scratch-random, scratch-active, transfer-random, and transfer-active paths, a scratch fit on each exact transfer-active subset, and a 600-label full-pool scratch reference. Paths started with 60 labels, acquired batches of 30, and reported budgets 60, 150, and 300. The same uncertainty/diversity rule used 24 informed and six exploratory selections for a full 30-recipe batch. Each round trained on all labels revealed so far, starting again from the original source checkpoint for transfer or fresh random weights for scratch. All layers could adapt. Target validation labels were reserved for cutoff selection, and were not used for acquisition or weight fitting.

Composition inputs were transformed analytically as $\left(x_{1:4}-0.2\right)/0.2$. The source network's first-layer weights and intercepts were transformed to preserve its predictions exactly in these coordinates. Adam state was reset at the start of each fit, retained across its training epochs, and the weights with the lowest observed training loss were restored. The documented training protocol used a 1,200-epoch limit, at least 100 epochs, patience of 60 epochs, and improvement tolerance 10^{-5}, with Adam learning rate 0.001, regularization 0.01, and batches of at most 64. Zero-shot and target-validation threshold-only controls kept model weights

fixed. In the retention experiments, the main source and target thresholded results both used the fitted model's target-validation cutoff; source-validation recalibration was a separate diagnostic.

M8. Replay eligibility and the two guided selection rules

For the no-replay comparison in Section 9, the saved P2 composition, P2 molecular, and P3 active-200 sources were compared after target-only fine-tuning on identical saved transfer-active subsets within each seed and budget (Methods M7). The completed P3 fits from the target-adaptation experiment were reused, and P2 fits started from their original source checkpoints on the same target examples. Network weights were updated without source replay; subsequent source-test evaluation measured retention. The subsequent duration and energy experiments focused on P2 molecular and P3 active-200. The P2 runs repeatedly adapted the same saved source committee under five adaptation seeds. P3 used a different previously acquired source committee for each seed. Replay candidates were restricted to original source-training recipes correctly classified by that source teacher at its original cutoff. For P3, candidates therefore came only from that seed's 200 acquired examples, not from the rest of the source pool. Source and target held-out examples were ineligible.

Each feasible policy selected 20 distinct examples once per source/seed and reused that memory at all three target budgets. Target-training identifiers were fixed to the saved transfer-active paths across replay policies within each seed and budget. Random-20 sampled uniformly without replacement. Slowest-20 selected the largest recorded stability-search durations; its matched random control reproduced its positive and negative counts exactly. The energy policies used predicted total mixing Gibbs energy, obtained by adding the analytical ideal contribution to the excess energy predicted by the frozen P1 Ridge model. Energy predictions selected memories and were not added as classifier features. Highest-20 and Lowest-20 retained the respective extremes, each with a class-matched random control. Ties in energy or duration were resolved by recipe identifier, and selected memories were put in canonical recipe order before training.

For the strict bridge, eligible recipes were ranked in descending predicted total mixing energy. Let j be the first negative label in this ranked list, using one-based indices. The positive side consisted of positions $j-10, \ldots, j-1$; the negative side consisted of the first ten negative-labeled recipes encountered from position j downward, skipping intervening positives. Feasibility required at least ten positives before the first negative and ten negatives from that anchor downward. If these requirements failed, the strict policy was unavailable: the anchor was not moved. Labels were previously obtained reference labels, not labels inferred from the energy surrogate. This rule samples a discrete contrast in the energy ordering and does not define a thermodynamic phase boundary.

The exploratory downstream rule instead collected the first ten positives encountered from the high-energy end, ignoring negatives encountered before the tenth positive. Starting

strictly after that tenth positive, it collected the next ten negatives. Earlier negatives were not reused. Its balanced control drew ten positives and ten negatives randomly from the same eligible pool. This changed rule enabled evaluation on all five P3 active-200 seeds; it is reported separately from the original strict experiment. The lower-energy transition suggested by the energy ordering was not tested. Strict-bridge results from the sole feasible P3 seed, 55, remain single-seed evidence and are compared with controls from that same seed.

M9. Replay training and computational accounting

Replay used the same target examples, source initialization, architecture, and target-validation procedure as the paired no-replay comparison. Molecular models kept their source-fitted feature scalers; paired scratch fits used the same fixed transforms. Composition models used the prediction-preserving coordinate conversion described in M7. These scratch controls isolate initialization under a shared representation and selected memory; they do not remove the source teacher or energy surrogate from the pipeline.

Inverse-frequency class weights were first computed on the concatenated target and memory labels. The target and memory portions were then rescaled separately to totals $0.8\,n_t$ and $0.2\,n_t$, where n_t is the number of target-training examples. Thus the data-loss allocation was

$$L_{data} = 0.8\,\frac{\sum_{i \in T} c_{y_i} \ell_i}{\sum_{i \in T} c_{y_i}} + 0.2\,\frac{\sum_{i \in M} c_{y_i} \ell_i}{\sum_{i \in M} c_{y_i}},$$

where c_{y_i} denotes the class weight from the combined supplied labels and ℓ_i is binary classification loss. These are domain-normalized class-weighted means; class weights were not recomputed independently within each domain. Total supplied weight remained n_t, preserving the full-dataset weight scale relative to the no-replay fit. Regularization was handled by the estimator, separately from this data-loss expression. Shuffled minibatches need not have an exact 80/20 allocation. This implements a fixed experience-replay tradeoff [8].

We distinguish reference-label counts, summed recorded reference durations, machine-learning fitting and acquisition durations, and incremental new fitting work. A reference-duration total sums saved per-recipe elapsed stability-search times over the selected set. Concurrent execution can make this sum differ substantially from campaign wall time; it is also not a measured CPU-hour total. In a retrospective benchmark, selecting fewer already-labeled recipes estimates a prospective acquisition saving rather than recovering time already spent. Model fitting, scoring, validation, evaluation, and source pretraining are reported or identified separately.

Training-label savings compare 200 with 1,191 source-training labels or 150 with 600 target-training labels. Including validation or evaluation changes the denominator, as shown in Table 7 and Figure 6. For an incremental target task, the previously trained source is an existing

asset; an end-to-end cost comparison must also include its construction. Replay reused source labels and required no new stability searches. The downstream extension’s 54 new committee fits and six reused control fits were distinguished explicitly: reused results carry zero incremental fitting cost, while 775.15 s is the sum of newly incurred fitting durations.

Appendix A. Model settings and implementation details

A1. Regression and classifier settings

The saved P1 run searched Ridge degrees 2, 3, and 4 with regularization parameters 0.0001, 0.01, and 1. Histogram gradient boosting used maximum leaf counts 7, 15, and 31 and learning rates 0.05 and 0.1, with 300 boosting iterations, minimum leaf size 10, and automatic early stopping disabled. The selected full-data Ridge was degree 4 with regularization 0.0001; boosting used 31 leaves and learning rate 0.1. Polynomial expansion was followed by training-only standardization and Ridge with an intercept. The saved run used eight worker processes with one numerical thread per worker.

P2 logistic regression searched $C \in \{0.1, 1, 10\}$, using L-BFGS, a 3,000-iteration cap, and tolerance 10^{-7}. RBF support-vector classification searched $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{scale, 0.1, 1\}$, with tolerance 10^{-5}. MLP settings crossed hidden layers (64,32), (128,64), or (256,128,64) with regularization 0.0001, 0.01, or 0.1. Every candidate family was considered with and without inverse-frequency class weighting. MLP training used ReLU, Adam, learning rate 0.001, batch size 64, a 1,200-iteration cap, tolerance 10^{-5}, and 40 iterations without sufficient training-loss improvement; internal validation-based early stopping was disabled. Final settings were chosen separately for each feature group and learner. The saved selection records specify the chosen settings for every feature group and learner.

The saved final P2 MLP selections used regularization 0.1 in every feature group. Composition, molecular, thermodynamic, and activity models selected hidden layers (128,64); the full-feature model selected (64,32). Only the molecular MLP selected inverse-frequency class weighting. These original source-fitting choices differ from the regularization 0.01 used in P3 and target adaptation, so comparisons between source families do not isolate training-set size alone.

The documented P3 protocol used a 1,200-epoch first attempt. If that attempt reached its limit with a convergence warning, the implementation allowed one warm-start continuation with a doubled cap on the same revealed examples. This stopping rule permits more than 1,200 total epochs when a continuation is triggered. No continuation passed information between acquisition rounds. The P4 implementation instead iterated partial-fit epochs under a fixed per-fit cap and restored the best training-loss weights. These implementation-level stopping rules are distinct from the realized epoch count of an individual fit.

A2. Dataset composition and experimental splits

P1 uses 17,121 training compositions for regression, whereas P2 uses 1,191 labeled training compositions for classification. In P3, each active-200 model is trained on its own set of 200 acquired compositions. For the replay experiments, candidate examples were drawn only from the training data of the corresponding source classifier.

The source validation and test sets contain 314 and 495 compositions, respectively, and remain unchanged throughout P2–P4. The target domain has separate validation and test sets of 200 compositions each. Its training, validation, and test splits contain 240/80/80 binary, 120/40/40 ternary, 60/20/20 quaternary, and 180/60/60 five-component compositions. All reported performance metrics are evaluated using these splits.

A3. Recorded software versions

The P1 and P2 experiments used NumPy 2.4.6, pandas 3.0.6, SciPy 1.17.1, scikit-learn 1.8.0, joblib 1.6.0, and Matplotlib 3.11.2. P2 additionally used threadpoolctl 3.7.0. The thermodynamic calculations used openCOSMO-RS_py (version 0.0.1, commit c2bc186).

Reference calculation times are reported as the sum of the individual recorded durations under the original computational conditions, as described in Methods M9.

Appendix B. Code and data availability

The code and supporting data for this study are available at:

https://github.com/qdms91/mixture-thermodynamics-ml

The repository contains the molecular inputs and calculated COSMO surfaces, thermodynamic datasets, reference instability labels, training and evaluation scripts, frozen experimental splits, selected trained models, and numerical results for P1–P4. It also provides a Conda environment specification and instructions for reproducing the calculations and machine-learning experiments.

Additional model checkpoints, detailed reference-search records, and prediction data will be provided as downloadable archives in the associated GitHub Release.

https://github.com/qdms91/mixture-thermodynamics-ml/releases/tag/v1.0.0

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